

Adventist University of Central Africa

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**NGO FUND PLATFORM: FUNDING
MANAGEMENT & OPERATIONS
SYSTEM**

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ABSTRACT

This project proposes the development of the NGO Fund Platform, a comprehensive digital system designed to enhance transparency, accountability, and operational efficiency within the Rwanda Paediatric Association (RPA). As a leading healthcare organization managing complex multi-donor portfolios from partners such as USAID, WHO, and UNICEF, RPA faces significant challenges in coordinating financial activities, ensuring compliance, and maintaining timely reporting. At present, the association relies heavily on fragmented manual processes, including disconnected Excel spreadsheets and physical receipts, which result in delays, increased administrative overhead, high risk of non-compliance with donor restrictions, and limited real-time visibility into the organization's financial health.

The primary objective of this project is to design and implement an integrated web-based system that centralizes RPA's core operational functions across ten critical modules, including Donor Management, Fund Collection, Expenditure Monitoring, Grant Allocation, and Automated Reporting. By unifying these modules under a single platform, the system will eliminate manual reconciliation points, provide real-time access to financial data, and support accurate, audit-ready reporting. The system will be developed using a modern technology stack comprising React for the frontend, Spring Boot for the backend, and PostgreSQL for database management, ensuring scalability, reliability, and security. The development process will follow the Agile methodology, allowing iterative design, continuous stakeholder feedback, and rapid adaptation to operational needs.

Beyond its application at RPA, the NGO Fund Platform is envisioned as a scalable digital model for other mid-sized Rwandan non-profits. Its adoption will contribute to the national agenda for digital transformation, smart administration, and professionalization of civil society organizations, providing a replicable framework that enhances operational efficiency, compliance, and strategic management across Rwanda's non-profit sector. By bridging the gap between proven digital management practices and local operational realities, the project aims to set a new standard for integrated, technology-enabled financial management in the country.

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INTRODUCTION

Background

The Rwanda Paediatric Association (RPA) is a non-profit professional organization dedicated to promoting quality pediatric services and enhancing evidence-based knowledge to improve child health outcomes in Rwanda. Established to unify pediatricians and child health workers, the RPA's mission encompasses fostering preventive, curative, palliative, and rehabilitative pediatric care. The association plays a pivotal role in national health strategy, actively collaborating with the Ministry of Health and the Rwanda Biomedical Centre (RBC) to develop national treatment guidelines and conduct Continuous Professional Development (CPD) training for healthcare providers (Rwanda Paediatric Association, 2023).

As the association expands its scope to manage complex grants from international partners (e.g., UNICEF, WHO) for initiatives like neonatal survival and malnutrition reduction, the volume of financial transactions has grown exponentially. However, the internal financial management systems have not evolved at the same pace. Currently, RPA relies on disjointed manual processes—physical payment vouchers and standalone Excel spreadsheets to manage these multi-donor funds. This gap between the organization's high-level clinical mission and its low-level administrative capacity creates significant operational risks, including delayed reporting and potential compliance findings. The operations of the RPA are supported by a complex and diverse portfolio of funding sources and strategic partnerships. The association collaborates with major international bodies such as USAID (e.g., Ingobyi Activity), the World Health Organization (WHO), UNICEF, and ENABEL.

In the non-profit sector, and particularly for an organization managing such multifaceted partnerships, transparent fund management is not merely an administrative task but a critical operational requirement. Each donor whether a government agency like USAID or a private foundation typically dictates specific reporting standards, restricted fund usage, and compliance metrics. The ability to accurately track, allocate, and report on these funds determines the organization's credibility and its ability to secure future funding. As the RPA continues to expand its scope, moving from capacity building to implementing large-scale health interventions, the need for a robust, transparent, and efficient system to manage these financial flows becomes paramount.

Project Justification

The growing inefficiencies and lack of digital integration within the Rwanda Paediatric Association's financial operations demand urgent attention. Although the non-profit sector in Rwanda plays a crucial role in national development, many organizations like RPA still depend on manual coordination and paper-based record management. RPA illustrates this challenge vividly: managing a multimillion-franc portfolio from diverse donors (USAID, WHO, UNICEF) using isolated spreadsheets and physical receipts. This heavy reliance on manual systems leads to reporting delays, inconsistencies in fund tracking, and a high risk of non-compliance. If left unaddressed, these inefficiencies threaten RPA's credibility with international partners, where transparency, real-time visibility, and audit readiness have become prerequisites for funding renewal.

The **NGO Fund Platform** directly aligns with both institutional and national development objectives. For RPA, the system supports its mission to promote quality pediatric services by ensuring that administrative resources are synchronized with program needs rather than wasted on manual reconciliation. On a national level, it advances **Rwanda's Vision 2050** and the **Smart Rwanda Master Plan**, which emphasize digital transformation and accountability across all sectors, including civil society. It also aligns with the national push for a cashless, paperless economy. By integrating donor management, fund tracking, and reporting into one digital platform, the project complements these policy frameworks and directly responds to stakeholder expectations for efficiency, accountability, and transparency in the health sector.

The tangible results of this system will be both measurable and impactful. The automation of reliable data entry and donor reporting is projected to improve financial data accuracy by over 90%, while reducing report generation cycles and administrative errors by at least 60%. Real-time analytics and financial dashboards will empower the executive committee to make informed, data-driven decisions regarding resource allocation. Furthermore, digital approval workflows embedded within the system will shorten the time between a project request and fund disbursement, ensuring that critical health interventions are not delayed by bureaucratic bottlenecks.

Beyond its immediate operational advantages, the project carries far-reaching socioeconomic implications for both the organization and the broader non-profit sector. Once successfully implemented at **RPA**, the NGO Fund Platform will serve as a scalable and replicable model for other medical associations, cooperatives, and non-governmental organizations across Rwanda. By standardizing financial management, donor reporting, and compliance processes, the platform will contribute to the strengthening of national accountability standards and governance practices. This enhanced capacity will enable local organizations to “graduate” from reliance on sub-granting arrangements to directly accessing funding from international donors, thereby increasing institutional autonomy and credibility.

In the long term, the system empowers organizations to manage larger and more complex donor-funded projects without a proportional increase in administrative costs or staffing requirements. Through automation and integrated financial controls, the platform reduces inefficiencies associated with manual processes, allowing resources to be redirected toward core programmatic activities. This operational scalability supports sustainable organizational growth and creates opportunities for high-value employment in areas such as non-profit financial management, compliance oversight, data analysis, and digital systems administration.

Empirical research and global evidence further underscore the necessity of this intervention. Studies conducted by organizations such as **NetHope** and **TechSoup** reveal that non-profits utilizing integrated digital financial management systems experience reductions of approximately **20–30% in administrative overhead** compared to organizations that rely on manual or fragmented processes. Despite these proven benefits, a substantial number of local NGOs continue to operate with paper-based or semi-digital systems, limiting their efficiency, transparency, and funding potential.

Comparative findings clearly indicate that digital transformation in fund management is not merely a technological enhancement but a strategic imperative for organizational sustainability and competitiveness. By adopting the NGO Fund Platform, **RPA** positions itself as a leader in operational excellence, accountability, and innovation within the non-profit sector. This strategic move strengthens the organization’s long-term sustainability and enhances its capacity to effectively deliver on its critical mission of improving child health outcomes while setting a benchmark for best practices across Rwanda’s NGO ecosystem.

Objectives

The objectives are divided into general and specific goals to ensure a clear direction and measurable outcomes.

General Objectives

The project enhances financial transparency and accountability through a digital platform that integrates donor data, restricted funds, and expenditure tracking for real-time monitoring and auditability. It improves operational efficiency and compliance by automating donor reporting, fund allocation, and expense approvals while ensuring adherence to USAID, WHO, and government requirements. The system supports data-driven strategic planning by providing analytical dashboards that offer insights into fund utilization, project burn rates, and donor engagement. Additionally, it contributes to Rwanda's Vision 2050 and the Smart Rwanda Master Plan by promoting digital innovation, paperless operations, and institutional capacity building.

Specific Objectives

- **Register and Manage Donor Profiles:** Develop a digital registration system to manage details of all institutional and private donors, including contact details, grant history, and specific reporting mandates.
- **Track Restricted Funds and Allocations:** Implement a real-time tracking module for monitoring specific grant budgets ("Restricted Funds") against project activities. This system aims to reduce reporting errors and "commingling" risks by at least 90%.
- **Facilitate Digital Approvals and Requisitions:** Create a secure internal communication channel for digital expense requisitions and approvals, shortening the time between a project request and fund disbursement by up to 50%.
- **Automate Compliance and Reporting Processes:** Digitize the creation, storage, and submission of critical financial reports, automatically generating formats required by major donors (e.g., USAID SF-425).
- **Enable Data Analytics for Financial Health:** Integrate analytical dashboards to evaluate organization-wide financial health, clearly visualizing "Funds Received" vs. "Funds Spent" to prevent under-spending or over-spending on grants.

Scope

The following sections define the precise boundaries of the **NGO Fund Platform** project, detailing what will be included in the final solution and what is considered out of scope.

Project Inclusion Scope The project focuses on the development and implementation of a centralized digital platform. The scope of the system includes the following core modules and functionalities:

1. Donor & Grant Management:

- Digital registration and management of donor profiles.
- Capturing and managing data on specific Grant Agreements, including timeframes and restriction rules.

2. Fund Collection & Income Tracking:

- Real-time recording of incoming funds from bank transfers and mobile money.
- Generation of digital receipts and acknowledgment letters for donors.

3. Project Allocation & Budgeting:

- Creation of project-specific budgets linked to master grants.
- Real-time visualization of "Budget vs. Actual" status for every project.

4. Expenditure & Requisition Module:

- Digital submission of purchase requisitions by project staff.
- Automated approval workflows for Finance Officers and the Executive Director.

5. Compliance & Reporting Dashboard:

- Digital creation and export of financial reports (PDF/Excel) tailored to donor requirements.
- Automated alerts for upcoming reporting deadlines or expiring grants.

6. Communication & Audit Logs:

- An integrated notification system for approval status updates.
- Immutable audit logs recording every user action for external auditors.

7. System Administration:

- Secure, role-based access control (e.g., Admin, Finance Officer, Project Manager, Auditor) to protect sensitive data.

Project Exclusion Scope To maintain a clear focus and ensure successful project completion, the following functionalities are explicitly excluded from the scope of this project:

- 1. Full Payroll Management:** The system will manage project-based stipends/consultancies but will not function as a full HR Payroll system for calculating statutory tax deductions (PAYE, RSSB) for permanent staff.
- 2. Clinical/Patient Data Management:** The platform tracks the *funds* used for medical projects (e.g., "Cost of medicines purchased") but will NOT store any patient medical records or clinical trial data.
- 3. Direct Bank Interface:** The system will record transactions based on bank statements but will not integrate directly with banking APIs to execute real-time money transfers.
- 4. Hardware Provisioning:** This is a software project. The scope does not include the procurement of computers, servers, or tablets for RPA staff.
- 5. Offline-First Functionality:** The system is designed as a cloud-based web application requiring internet connectivity; it is not intended for fully offline usage in remote areas without synchronization.

LITERATURE REVIEW

Introduction

Digital technologies are transforming the non-profit sector worldwide by enabling better fund management, compliance tracking, and data-driven decision-making. In Rwanda, the civil society sector has experienced significant growth in funding complexity, highlighting both the potential and operational constraints of current financial administration practices. Digital platforms that centralize donor data, link expenditure to specific grant restrictions, and provide real-time audit visibility are now widely recognized as essential enablers for meeting stringent international donor requirements (e.g., USAID, EU, World Bank). These digital interventions range from simple accounting software to full Enterprise Resource Planning (ERP) systems that support compliance standards and rapid response to audit queries.

Review of Existing Works

- **Evolution of Digital Financial Management Systems:** Over the past decade, financial management platforms have evolved from basic bookkeeping tools into comprehensive digital ecosystems integrating donors, implementers, and auditors. Early solutions such as QuickBooks and Sage focused on general ledger accounting, while newer platforms like Salesforce Non-Profit Cloud, NetSuite, and Microsoft Dynamics 365 now combine CRM, project management, and fund accounting. Research by *NetHope* (2022) shows that digital financial platforms have increased donor reporting accuracy by 60% and reduced manual reconciliation errors by 45% in participating NGOs.
- **Donor Profiling and Compliance TECHNOLOGIES:** Digital donor management has become a key enabler of funding sustainability. Platforms like Bloomerang and generic CRMs enable NGOs to digitally register donors, track engagement history, and generate audit-ready reports for restricted funds. Case studies from Kenya's NGO sector and Uganda's Civil Society Fund confirm similar outcomes, improving transparency and ensuring fund separation throughout the project lifecycle. These systems also enhance coordination between finance and program teams by enabling budget visibility and burn-rate forecasting.
- **Grant Reporting and Documentation Systems:** Automation of reporting processes has reduced the administrative burden in international aid management. Specialized tools

such as Fluxx, Amp Impact, and GrantHub integrate budget tracking, digital receipts, and narrative reporting within centralized databases. According to *TechSoup* (2023), organizations using digital grant management systems report a 50% reduction in reporting time and 30% lower compliance errors. However, most regional professional associations, including those in Rwanda, still rely on fragmented spreadsheets and manual approvals, increasing turnaround time and audit risks.

- **Audit readiness and strong internal controls** are critical requirements for organizations such as RPA, particularly because their funding partners demand strict compliance with frameworks such as the **United States Government (USG) Standard Provisions** and comparable international regulatory standards. The adoption of digital compliance tools has significantly transformed audit preparation and execution by enabling finance teams to securely upload and link supporting documentation—including receipts, contracts, approvals, and payment records—directly to individual financial transactions. This centralized and traceable approach improves transparency, reduces the risk of missing documentation, and strengthens internal control mechanisms. According to a 2022 study by the Chartered Institute of Public Finance and Accountancy (CIPFA), NGOs that implemented digital audit trails achieved audit completion rates up to 35% faster than organizations
- Similarly, the use of data analytics and decision-support platforms is increasingly recognized as a strategic asset in modern NGO financial management. As organizations face growing pressure to optimize limited resources, ensure accurate cash flow forecasting, and demonstrate measurable impact, data-driven management approaches are becoming essential. Tools such as Tableau for Nonprofits and Microsoft Power BI leverage cloud-based analytics to generate real-time dashboards, trend analyses, and predictive financial models that support informed decision-making. Empirical evidence from USAID pilot programs in East Africa (2021–2023) indicates that partner organizations utilizing analytics-enabled systems improved budget utilization forecasting accuracy by 28% and reduced disallowed costs by 20%. These findings demonstrate that analytics platforms not only enhance financial oversight but also contribute to stronger compliance, improved donor confidence, and more effective program delivery.

Identification of Gaps

- **Integration and End-to-End Orchestration Gap:** Many existing financial management solutions tend to focus on isolated segments of the organizational workflow, such as accounting, customer relationship management (CRM), or project management, without providing fully integrated end-to-end processes. This results in fragmented systems where donor profiles, specific restricted grants, expense approvals, and final reporting are not seamlessly connected. Such gaps necessitate manual handovers and reconciliations between modules like the "Donor CRM" and the "Accounting Ledger," reducing the efficiency and reliability of digitization efforts and increasing the risk of errors and delays in financial reporting.
- **Cost and Complexity for Mid-Sized NGOs:** While enterprise-grade ERP systems such as **SAP** and **NetSuite** provide comprehensive financial and operational features, their high licensing costs, technical complexity, and significant implementation requirements make them unsuitable for mid-sized organizations like RPA. These systems often require specialized IT support and extensive training, imposing financial and operational burdens that smaller NGOs cannot sustain. On the other hand, simpler tools such as **Excel** or **QuickBooks** are affordable and widely used but lack the specialized "Fund Accounting" logic necessary to manage multi-donor restricted funds, ensure compliance with donor regulations, or generate audit-ready reports without expensive and time-consuming customization. This leaves mid-sized NGOs caught between overly complex solutions and insufficient basic tools, creating a critical gap in accessible, scalable financial management solution
- **Context-Specific Compliance Logic:** Generic international software often lacks the specific taxation and statutory logic required in the Rwandan context (e.g., specific EBM receipt handling, withholding tax calculations). There is a need for solutions that align with local regulatory frameworks while meeting international standards.
- **User-Centric Design for Non-Finance Staff:** Most financial systems are designed for accountants. Literature indicates a shortage of platforms that are user-friendly for *program managers* and *executive directors*, who need simple dashboards to approve expenses and view budget status without navigating complex accounting interfaces.

Summary

The literature underscores a strong rationale for integrated digital platforms that unify donor management, fund accounting, expense approval workflows, and automated reporting to meet international compliance standards while enhancing operational efficiency. Empirical evidence from regional studies and reports by international bodies demonstrates not only the technical feasibility of such systems but also the measurable benefits, including reduced administrative overhead, faster audit readiness, improved budget tracking, and enhanced transparency. Despite these proven advantages, significant gaps remain that hinder the adoption and full realization of value from digital financial management solutions. Key barriers include limited end-to-end integration, prohibitive cost and complexity for mid-sized organizations, insufficient adaptation to local regulatory and taxation requirements, and low usability for non-finance staff, which collectively restrict widespread uptake.

The proposed NGO Fund Platform directly addresses these challenges by providing a unified, contextually tailored solution designed to reflect RPA's operational realities and the broader professionalization goals of Rwanda's civil society sector. By combining the functions of donor relationship management, restricted fund accounting, expense approvals, and reporting into a single platform, the system minimizes manual handovers, reduces errors, and supports compliance with both national and international standards. Furthermore, its design emphasizes user-friendly interfaces suitable for program managers, executive directors, and finance teams alike, ensuring broad adoption across organizational roles.

This strategic focus positions the NGO Fund Platform not merely as a technological upgrade, but as a scalable and sustainable solution capable of translating globally validated digital management practices into a locally relevant, practical platform. In doing so, it contributes to the long-term strengthening of financial governance, operational transparency, and professional capacity within Rwanda's non-profit ecosystem, while enabling RPA to handle increasingly complex donor-funded projects with efficiency, accountability, and scalability.

Proposed Solution

Overview of the Solution

Brief Description of the Solution The **NGO Fund Platform** is a cloud-based digital management system designed to modernize and unify the financial and operational workflows of the Rwanda Paediatric Association (RPA). The platform provides an integrated ecosystem that connects donors, finance officers, project managers, and auditors through a centralized web interface. The solution automates manual workflows such as donor registration, fund allocation, expenditure reporting, and compliance monitoring, ensuring real-time visibility and data synchronization across the entire grant lifecycle. By leveraging restricted-fund accounting logic and audit trails, the system enhances transparency, reduces administrative inefficiencies, and improves compliance with international donor standards such as USAID Standard Provisions.

Core Concept of Methodology Behind the Solution The methodology behind the NGO Fund Platform is based on a "**Digitally Integrated Financial Compliance Model (DIFCM)**" that emphasizes data integrity, real-time auditability, and process automation. This model transforms fragmented manual processes into a unified digital ecosystem by applying the following core components:

1. **Workflow Digitization and Automation:** Converts paper-based requisition forms and physical receipts into digital workflows, enabling automatic validation and seamless task tracking between the finance and program departments.
2. **Data Integration and Centralized Architecture:** Establishes a secure, cloud-hosted data repository serving as a single source of truth for donor agreements, budget lines, bank transactions, and audit logs. This reduces redundancy and enhances data consistency across all stakeholders.
3. **Real-Time Restricted Fund Accounting:** Employs rigorous validation logic to strict regulations, ensuring that "Restricted Funds" cannot be accidentally allocated to unapproved budget lines, guaranteeing compliance by design.
4. **Analytics and Financial Insights:** Integrates data analytics dashboards to provide the Executive Committee with insights on burn rates, budget variances, and funding sustainability. Predictive models will support cash flow forecasting.

5. **Role-Based Access Control (RBAC):** Adopts an intuitive interface tailored to different user roles—Finance Director, Project Coordinator, and External Auditor—ensuring appropriate data visibility and segregation of duties.

How the Solution Ties Back to the Problem The RPA currently faces multiple operational challenges, including inefficient coordination with diverse international donors, reliance on manual Excel spreadsheets, fragmented data systems, and high compliance risks. These inefficiencies contribute to delayed reporting, audit vulnerabilities, and the potential loss of funding. The **NGO Fund Platform** directly addresses these pain points by:

- **Digitizing Fund Tracking:** Enabling real-time visibility of incoming grants, budget allocations, and remaining balances.
- **Automating Donor Reporting:** Generating financial reports (e.g., SF-425) digitally to reduce administrative delays and human error.
- **Enhancing Audit Traceability:** Linking each expenditure to its origin grant, approved requisition, and digital receipt for full financial transparency.
- **Strengthening Compliance Management:** Integrating compliance checklists to help RPA maintain donor standards with automated alerts for ineligible expenses.

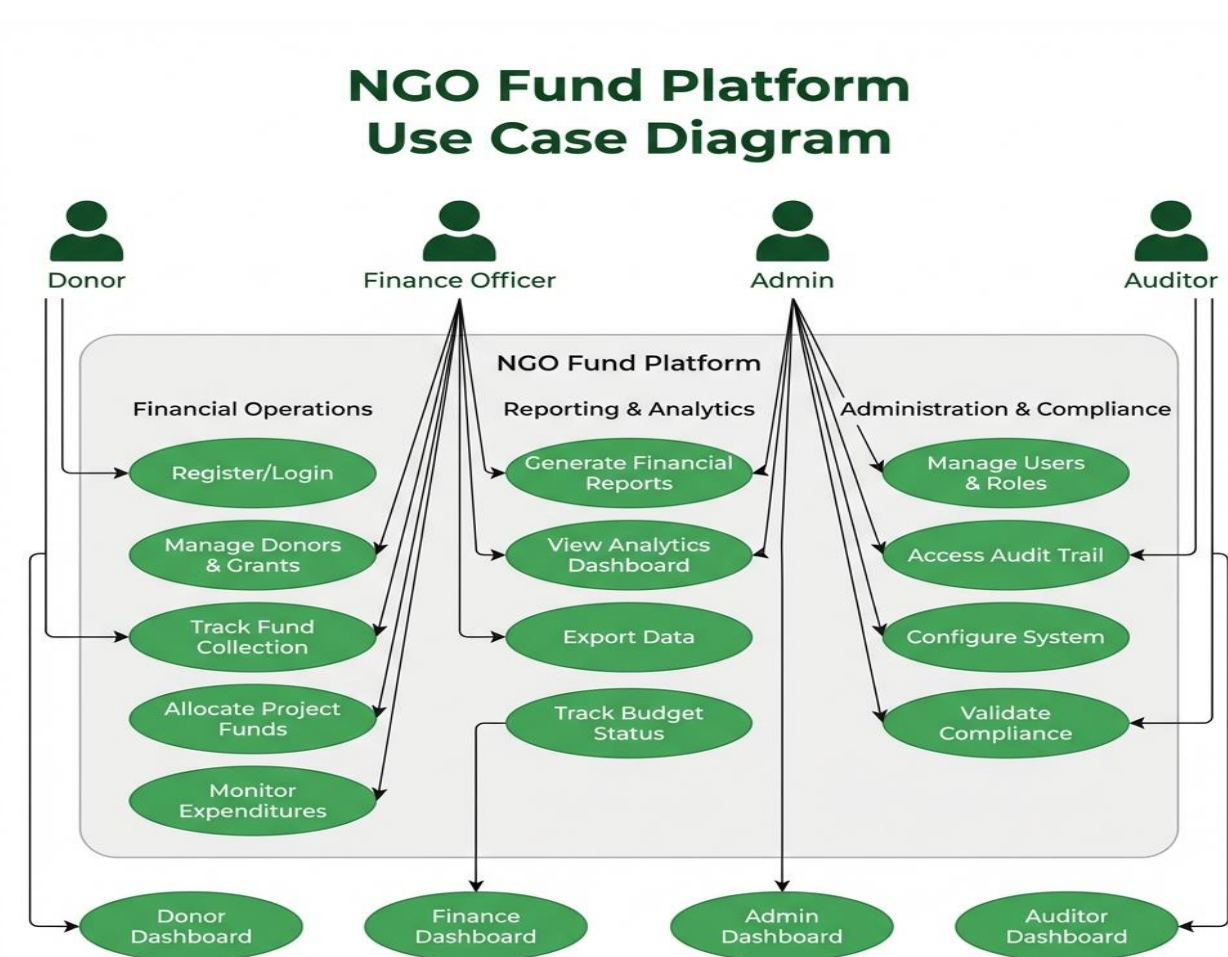
By bridging these operational gaps, the platform will transform RPA's manual, fragmented workflows into an efficient, transparent, and data-driven digital ecosystem, positioning the association as a model of financial integrity in Rwanda's civil society sector.

System Design

The NGO Fund Platform is tailored for the high-stakes environment of non-profit financial management. Built on a robust **Three-Tier Web Architecture**, the platform centralizes operations into a secure web-based hub.

1. **Data Collection and Validation Layer:** This layer is responsible for capturing financial data, including donor grants, bank receipts, and expense requests. It includes tools for data normalization (e.g., currency conversion), validation algorithms to prevent "commingling" of funds, and bulk import tools for bank statements.
2. **Compliance & Accounting Engine:** This core module applies "Fund Accounting" logic. It ensures that every transaction is tagged with a Donor ID and Project ID. The system flags potential compliance issues, such as budget overruns or ineligible expense categories (e.g., alcohol, entertainment) before they are processed.

3. **Reporting & Analytics Module:** This module analyzes financial data to generate donor-specific reports. It provides real-time insights into "Burn Rates" (how fast money is being spent) and generates the "Statement of Activity" required for annual audits.
4. **Privacy and Security Layer:** This layer ensures that all sensitive financial and donor data is encrypted and compliant with the Law on Protection of Personal Data and Privacy. It includes secure object storage for digital contracts and strict Role-Based Access Controls (RBAC).
5. **User Interface (UI) Dashboard:** A centralized Web-Based Interface allows all stakeholders to interact with the system. Users can view their specific KPIs: Project



Managers see their budget status, while Auditors see read-only transaction logs.

Figure 1: Use Case Diagram

Key Features

The system is organized into ten distinct yet integrated modules, each designed with specific features to address the core problem of disjointed financial management.

1. Donor Management Module

- **Description:** A comprehensive CRM for managing donor relationships.
- **Key UI Elements:** Donor registration form, contribution history table, and communication logs.
- **Features:** Automated acknowledgment messages, donor categorization (e.g., USAID, EU, Private), and engagement tracking to ensure funding sustainability.

2. Fund Collection & Tracking Module

- **Description:** The entry point for all incoming revenue.
- **Key UI Elements:** Fund receipt entry form, real-time incoming funds dashboard, and bank reconciliation interface.
- **Features:** Multi-currency support (USD/EUR/RWF) with automatic conversion, real-time alerts for incoming transfers, and separation of restricted vs. unrestricted incoming funds.

3. Project Fund Allocation Module

- **Description:** The "brain" of the restricted fund compliance system.
- **Key UI Elements:** Project budget dashboard, allocation form (linking specific grants to specific projects), and disbursement timeline.
- **Features:** Budget adherence monitoring (preventing allocation > available grant), reallocation approval workflows, and historical logs of all fund movements.

4. Expenditure Monitoring Module

- **Description:** Examples of strict internal controls on spending.
- **Key UI Elements:** Expense entry form (with mandatory category selection), approval workflow interface, and digital receipt upload.
- **Features:** Real-time "Budget vs. Actual" validation during entry, digital audit trail visibility, and expense categorization filters (e.g., Travel, Per Diem, Supplies).

5. Reporting & Analytics Module

- **Description:** The output engine for donor and internal reporting.
- **Key UI Elements:** Custom report builder (drag-and-drop), dashboard with KPI cards (Burn Rate, Funding Gap), and "One-Click" export to PDF/Excel.
- **Features:** Generates standard reports (Statement of Activity, Balance Sheet) and donor-specific formats (e.g., USAID SF-425) automatically.

6. Staff & Operations Module

- **Description:** Manages the human element of project implementation.
- **Key UI Elements:** Requirement capture forms, process documentation editor, and staff feedback submission area.
- **Features:** Requirement validation tracking and collaborative workflows to ensure the system evolves with staff needs.

7. System Testing & Validation Module

- **Description:** Ensures reliability and user acceptance.
- **Key UI Elements:** Bug tracking board, UAT feedback forms, and test case management tables.
- **Features:** Structured tracking of validation results to ensure audit readiness before full deployment.

8. User & Access Management Module

- **Description:** The security backbone of the system.
- **Key UI Elements:** Role assignment interface, permission matrix editor, and login activity logs.
- **Features:** Granular Role-Based Access Control (RBAC) separating "Creator" from "Approver," and session timeout settings for security compliance.

9. Dashboard & Transparency Portal

- **Description:** High-level visibility for the Executive Committee.
- **Key UI Elements:** Real-time financial health widgets, transparency scorecard, and project funding heat maps.

METHODOLOGY

Development Approach

For the development of the **NGO Fund Platform**, we will adopt an **Agile Scrum** development methodology. Agile is a flexible, iterative approach that emphasizes collaboration, rapid prototyping, and continuous improvement. Given the complex and evolving nature of the non-profit financial sector, which involves multiple stakeholders (donors, finance officers, project managers, auditors), Agile offers a suitable framework for the successful development and deployment of the platform.

Detailed Description of the Agile Methodology Agile focuses on incremental development through iterative cycles known as **Sprints**. Each sprint typically lasts between 2 and 4 weeks, during which specific features or modules of the platform (e.g., the Donor Management Module or Expenditure Reporting) are developed, tested, and refined. At the end of each sprint, a functional increment of the platform is delivered, with feedback from RPA's Executive Committee integrated into subsequent cycles.

The key principles of our approach include:

- **Collaboration Over Contract Negotiation:** Frequent communication with RPA management and finance staff ensures the project stays aligned with real-world operational needs and donor compliance changes.
- **User-Centric Focus:** The project prioritizes the needs of the end-users, specifically the ease of data entry for non-financial program staff and the clarity of dashboards for executive decision-makers.
- **Iterative and Incremental Development:** We will build the system module by module (e.g., starting with Grant Tracking, then moving to Requisitions), ensuring critical features are tested early.
- **Flexibility to Change:** As donor regulations or tax laws evolve, Agile allows us to adapt the system without disrupting the entire project structure.

Suitability of Agile for the Project Agile is uniquely well-suited for the development of the NGO Fund Platform due to the complex and evolving nature of the non-profit sector. The

requirements for compliance and reporting are not static; they change frequently with new donor agreements and government tax regulations. An Agile approach allows the development team to adapt quickly to these regulatory shifts without derailing the entire project, ensuring that the final product remains compliant at launch.

Furthermore, the project involves high-stakes financial logic that requires rapid prototyping and continuous testing. By breaking the complex fund accounting rules into smaller, manageable increments (e.g., handling "Income" separately from "Expenditure"), the team can identify integration risks and logic errors early in the process. This iterative cycle, coupled with a strong user-centric focus, ensures that the system interfaces—designed for non-finance program staff—are intuitive and effective, minimizing the risk of user rejection during the final rollout.

Stages of the Agile Process for This Project

1. **Sprint Planning:** At the start of each cycle, we will outline the goals based on priorities. A sprint might focus on a key milestone like "Implementing the Restricted Fund Logic" or "Building the SF-425 Report Generator".
2. **Sprint Execution:** The development team will implement the required features. This involves coding the backend logic for fund accounting, designing the frontend dashboards, and integrating the audit logging tools.
3. **Daily Standups:** Short meetings to track progress and resolve blocking issues immediately, ensuring coordination between the database, backend, and frontend tasks.
4. **Testing and Validation:** Regular testing will be conducted, including unit tests for individual modules (e.g., testing the currency conversion) and user acceptance testing (UAT) to ensure the workflow matches the physical approval process.
5. **Sprint Review and Retrospective:** At the end of each sprint, the team will demonstrate the new features to stakeholders to gather feedback and identify areas for improvement before moving to the next module.
6. **Deployment:** Once the core modules are refined, the system will be deployed in a controlled pilot phase for a specific grant or project before full-scale rollout.

Tools and Technologies

The selection of the technology stack for the NGO Fund Platform is driven by the need for high performance, data security, and scalability. The system utilizes a modern, microservices-ready architecture to ensure it can handle the complex workflows of grant management.

Front-end Development The frontend will be built using **React.js**, a powerful JavaScript library chosen for its component-based architecture which enables responsive and interactive user interfaces. To ensure design consistency and responsiveness across devices, we will utilize **Tailwind CSS**. For complex state management, such as real-time budget updates, **Redux** will be employed. Additionally, **Recharts** will be used to render the critical financial visualizations and analytics dashboards.

Backend Development The core application logic will be developed in **Java** using the **Spring Boot** framework. This combination is selected for its enterprise-grade stability, strict type safety, and robust security features, which are essential for financial systems. **Spring Security (OAuth 2.0)** will manage authentication and enforce role-based access controls, while **Hibernate (JPA)** will handle secure and efficient communication with the database.

Database Systems To ensure absolute data integrity, we will use **PostgreSQL**, an open-source relational database known for its ACID compliance. This ensures that financial transactions are processed reliably without data loss. Unstructured data, such as scanned receipts and contracts, will be stored securely in **Cloud Object Storage (e.g., AWS S3)** to maintain database performance.

AI and Machine Learning For the predictive analytics components, **Python** will be used due to its extensive data science ecosystem. Libraries like **Scikit-learn** and **Pandas** will be utilized to clean historical data and build models for cash flow forecasting and burn rate analysis.

DevOps and Security The application will be containerized using **Docker** to guarantee consistency across environments and deployed via **GitHub Actions** for continuous integration. Security is paramount; we will implement **Spring Security** with **JWT** for stateless authentication, **SSL/TLS** for encryption in transit, and **Bcrypt** for password hashing.

Data Collection

For the NGO Fund Platform, accurate and timely data collection is fundamental to ensuring end-to-end traceability, compliance with donor standards, and efficient financial planning. This section outlines the specific data points required, derived from current manual ledgers, and the methods used to digitize their capture.

1. Donor & Grant Data (Source of Funds) To establish financial traceability, the system will capture data currently recorded in physical grant agreements. This includes **Donor Details** (Organization Name, Contact Person), **Grant Specifics** (Agreement Number, Award Amount, Currency), and the specific **Restriction Logic** that dictates allowable costs.

2. Expenditure & Transaction Data Data generated during project implementation is critical for burn-rate analysis. We will collect **Requisition Metrics** (Request Date, Project Link), **Approval Logs** (Approver ID, Timestamp), and precise **Transaction Details** (Vendor Name, Invoice Number, Payment Method).

3. Compliance & Audit Data For internal controls, the system will capture **Supporting Documents** (Scanned Receipts, Timesheets) and immutable **Audit Trails** (User IDs, timestamps of all modifications) to provide a "proof of execution" for every expense.

Data Collection Methods We will employ **Web-Based Entry Forms** for day-to-day activities like expense requisitions and grant tracking. For historical or bulk data, a **Bulk Data Ingestion** tool will parse standardized .csv files from bank portals. The system will also create **System-Generated Logs**, automatically assigning unique Transaction IDs to eliminate manual errors.

Data Usage and Security All data collection adheres to strict privacy standards. **Data Privacy** is ensured by anonymizing PII for reporting. **Data Encryption** (SSL/TLS and AES-256) protects records in transit and at rest. **Role-Based Access Control** limits visibility based on the user's function, and the system is fully compliant with **Rwanda's Data Protection Law**.

Testing and Validation

To ensure that the NGO Fund Platform is robust and meets critical operational needs, a rigorous testing framework will be implemented.

Testing Strategy Our strategy involves a multi-layered approach. **Unit Testing** will be conducted first on individual components, such as currency conversion logic and tax formulas, to ensure they function correctly in isolation. Following this, **Integration Testing** will verify that the Spring Boot API, PostgreSQL database, and Web Dashboard interact seamlessly, ensuring that financial transactions update all relevant ledgers simultaneously. Finally, **Model Testing** will validate the accuracy of predictive analytics using historical spending data.

Validation Against Requirements Validation will focus on three key areas: **Accuracy**, **Real-World Usability**, and **Compliance**. Financial accuracy will be validated against historical manual audits to ensure zero computational errors. Usability will be tested through **Pilot Programs** where the finance team runs the system in parallel with current methods to identify discrepancies. Regulatory compliance will be verified against USAID reporting standards and local tax laws to ensure the system produces valid legal documents.

Tools and Frameworks We will utilize **JUnit** for automated backend testing and **Jest/Cypress** for frontend user flow verification. **User Acceptance Testing (UAT)** will involve finance officers simulating real-world scenarios to validate operational efficiency. **Error Metrics** will be used to quantify the accuracy of financial forecasts.

Continuous Validation Validation is an ongoing process. We will implement continuous monitoring to update the system as tax rates or donor templates change, ensuring the platform remains compliant and accurate throughout its lifecycle. verify the system against real-world scenarios.

WORK PLAN AND TIMELINE

Tasks and Milestones

The **NGO Fund Platform** project will be broken down into manageable tasks and milestones using the Work Breakdown Structure (WBS). Each task is formulated using the **SMART** (Specific, Measurable, Achievable, Relevant, Time-bound) criteria to ensure clear, actionable, and measurable outcomes.

1. Project Initiation and Planning

- **Task 1.1: Define project scope and objectives.**
 - **SMART Goal:** Clearly outline the NGO Fund Platform's scope and specific goals within the first week. This will include detailed problem statements regarding fund tracking inefficiencies and compliance gaps.
 - **Milestone:** Project scope and objectives document completed.
- **Task 1.2: Identify stakeholders and form the project team.**
 - **SMART Goal:** Complete stakeholder mapping (RPA Executive Committee, Finance Officers, Donor Representatives) within the first two weeks.
 - **Milestone:** Project team and stakeholders confirmed.

2. Requirements Gathering and Analysis

- **Task 2.1: Conduct interviews with key stakeholders.**
 - **SMART Goal:** Complete interviews with at least 5 key stakeholders (Finance Director, Project Coordinators, External Auditor) by Week 4 to understand reporting needs and workflow bottlenecks.
 - **Milestone:** Requirements document drafted based on interviews.
- **Task 2.2: Analyze existing financial systems.**
 - **SMART Goal:** Complete the analysis of at least three similar grant management or fund accounting systems (e.g., QuickBooks, Salesforce Non-Profit) by Week 5.
 - **Milestone:** Comparative analysis report completed.

3. Solution Design and Development

- **Task 3.1: Design system architecture and database schema.**
 - **SMART Goal:** Complete the design of the Three-Tier Web Architecture and the PostgreSQL relational database schema (Donors, Grants, Expenditures) within 6 weeks.
 - **Milestone:** Architecture design document approved.
- **Task 3.2: Develop the User Interface (UI).**
 - **SMART Goal:** Develop the high-fidelity prototype for the Finance Administration Dashboard (React.js) by Week 10.
 - **Milestone:** UI prototype and interface demo complete.

4. Data Collection and Preprocessing

- **Task 4.1: Identify and collect historical data.**
 - **SMART Goal:** Gather at least 80% of the required historical data (past grant agreements, bank statements, audit reports) from RPA archives by Week 8.
 - **Milestone:** Data collection and initial digitization complete.
- **Task 4.2: Preprocess data for Analytics Engine.**
 - **SMART Goal:** Complete cleaning and normalization of 100% of the collected financial data by Week 9 to ensure it is ready for the burn-rate models.
 - **Milestone:** Cleaned and pre-processed dataset ready for analysis.

5. Analytics Engine Development

- **Task 5.1: Develop predictive models.**
 - **SMART Goal:** Develop the Scikit-learn regression models for **Cash Flow Forecasting** and **Budget Burn Rate Prediction** within 3 weeks.
 - **Milestone:** Predictive models trained and ready for testing.
- **Task 5.2: Validate model accuracy.**
 - **SMART Goal:** Achieve a forecast accuracy of at least 95% on historical audit data by Week 13 (financial accuracy is critical).
 - **Milestone:** Final analytics engine validated.
 -

6. Integration and Deployment

- **Task 6.1: Integrate Backend and Frontend.**
 - **SMART Goal:** Complete the integration of the Spring Boot API with the React Dashboard within 2 weeks of core development.
 - **Milestone:** Integration test completed.
- **Task 6.2: Deploy solution to Pilot Server.**
 - **SMART Goal:** Deploy the full web platform to a cloud staging environment (e.g., AWS/DigitalOcean) by Week 15.
 - **Milestone:** System deployed and operational.

7. Testing, Validation, and Feedback Collection

- **Task 7.1: Conduct User Acceptance Testing (UAT).**
 - **SMART Goal:** Complete system testing within 2 weeks after deployment, where Finance Officers simulate real grant allocation and expense reporting scenarios.
 - **Milestone:** Successful validation and feedback gathered.
- **Task 7.2: Refine system based on feedback.**
 - **SMART Goal:** Implement critical UI/UX improvements and bug fixes based on UAT feedback by Week 17.
 - **Milestone:** Final system iteration completed.

8. Final Review and Project Closure

- **Task 8.1: Prepare project documentation and final report.**
 - **SMART Goal:** Complete the final technical documentation and user manuals by Week 18.
 - **Milestone:** Final project report submitted.
- **Task 8.2: Conduct project review meeting.**
 - **SMART Goal:** Hold the final review meeting with university supervisors and RPA stakeholders within 1 week of project completion.
 - **Milestone:** Project formally closed.

Timeline (Gantt Chart)

The Gantt chart below provides a comprehensive visual timeline of the major tasks and milestones throughout the project lifecycle. Spanning 18 weeks, it delineates each key activity, from initiation to final review, and highlights how phases overlap, interact, and build on one another. This chart plays a vital role in strategic planning and execution by:

- Offering a clear roadmap for when specific deliverables should be completed.
- Showing dependencies between tasks, such as how system testing relies on a completed prototype.
- Helping the project team allocate time and resources efficiently.
- Facilitating project monitoring, enabling early identification of delays or bottlenecks.

Each task is aligned with a defined start and end date, and the Gantt chart is synchronized with the SMART-based Work Breakdown Structure (WBS). It is a living tool, meant to evolve with the project and serve as both a guide and a diagnostic resource during implementation.

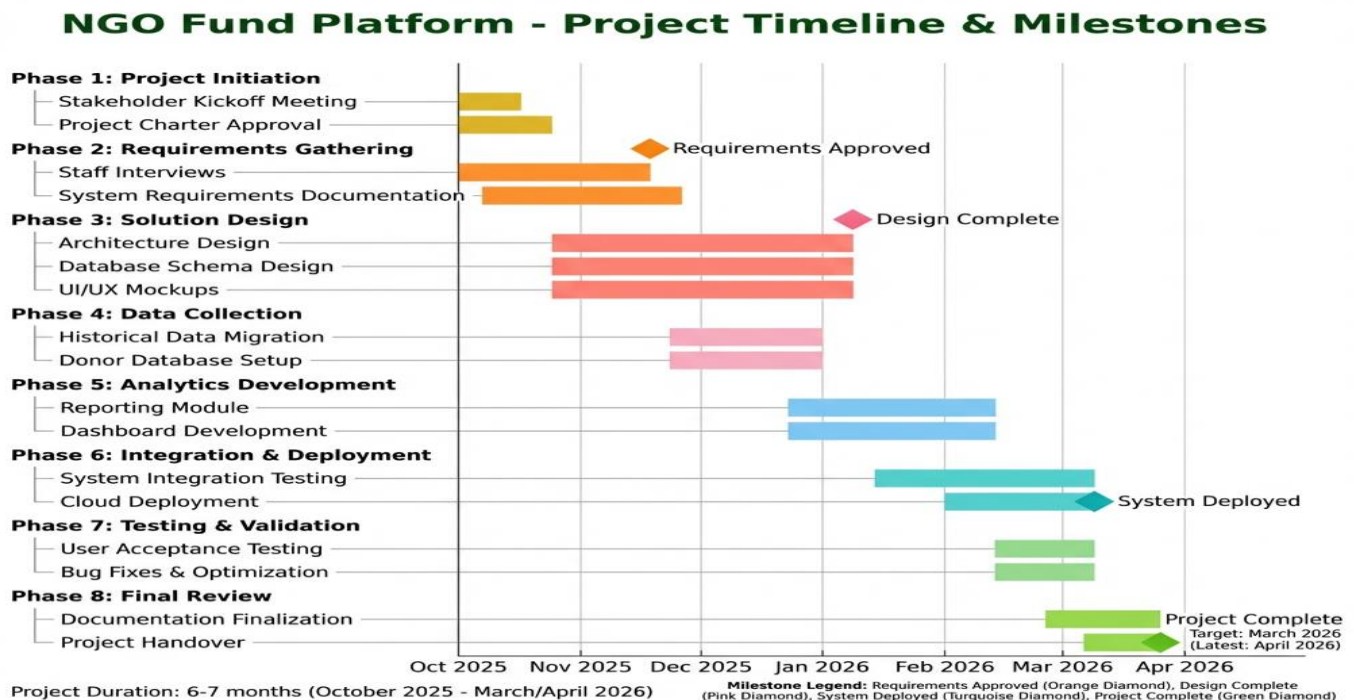


Figure 2: Project Gantt Chart

EXPECTED OUTCOMES

The **NGO Fund Platform** is expected to deliver tangible and measurable outcomes that contribute to modernizing the financial management practices of the Rwanda Paediatric Association. These outcomes will reflect both the functional deliverables of the software and the broader impact on operational efficiency, donor compliance, and institutional sustainability.

Final Deliverables

The project will produce the following deliverables:

1. **Enhanced Financial Traceability and Compliance:** By leveraging end-to-end fund tracking and automated compliance management, the system will significantly boost audit readiness. Real-time monitoring of funds from the bank receipt to the final expense ensures that every transaction meets international donor standards (such as USAID Standard Provisions), reducing the risk of disallowed costs and enhancing the reputation of RPA.
2. **Real-Time Operational Visibility and Efficiency:** RPA management and its partners will benefit from real-time access to web-based dashboards that track key metrics such as budget burn rates, grant balances, and expense statuses. This data-driven approach will support evidence-based planning, reduce administrative delays through better cash flow scheduling, and optimize the utilization of restricted funds.
3. **Enhanced Coordination Between Stakeholders:** The NGO Fund Platform will serve as a centralized digital hub where Finance Officers, Project Managers, and Executive Directors can interact seamlessly. This integration will reduce the current fragmentation in the financial workflow, streamline communications regarding budget reallocations, and improve overall efficiency in the reporting process without relying on dispersed Excel files.
4. **AI-Driven Predictive Insights:** With its integrated machine learning engine, the platform transforms historical spending data into actionable forecasts for cash flow and burn rates. This forward-looking capability allows RPA to move from reactive crisis management to proactive financial planning, optimizing resource allocation and ensuring that grant objectives are met with financial precision.

5. **Support for National Digitalization Goals:** By strengthening the civil society value chain and providing a robust digital infrastructure for fund management, the platform aligns directly with the government's strategic objectives and **Smart Rwanda** initiatives. It supports the goal of professionalizing the non-profit sector by modernizing the operational capabilities of associations, ensuring they can attract and retain international funding.
6. **Scalable and Adaptable Platform:** The system's modular structure and cloud-based architecture make it highly scalable, enabling expansion beyond RPA. The platform can be easily replicated and customized for use by other NGOs or professional associations across Rwanda, maximizing reviewing its impact on the national development sector.

Potential Limitations

While the NGO Fund Platform offers a transformative solution for RPA, several potential limitations and implementation challenges must be acknowledged and managed to ensure project success.

1. **Data Availability and Quality for AI:** The effectiveness of the Predictive Analytics Engine relies heavily on the volume and accuracy of historical data. Since RPA currently relies on manual spreadsheets and paper receipts, the initial historical data may require significant cleaning and normalization before it can be used to train the machine learning models. Consequently, the accuracy of cash flow forecasts may be lower during the initial months of operation until a sufficient volume of high-quality, standardized data is collected by the new system.
2. **Internet Connectivity and Infrastructure:** As a cloud-based Web Dashboard, the system requires a stable internet connection to function effectively. While the headquarters in Kigali may have reliable access, field coordinators operating in rural provinces may experience connectivity fluctuations. This could temporarily hinder real-time data entry for expense requisitions, potentially delaying the synchronization of field expenses with the central database.
3. **User Adoption and Change Management:** Transitioning from a familiar, albeit inefficient, Excel-based workflow to a fully digital ecosystem represents a significant cultural shift for staff. Finance officers and project managers accustomed to manual

vouchers may face a learning curve or resistance to adopting the new interface. Ensuring consistent user engagement will require comprehensive training programs and a period of parallel operation ("shadow run") to build trust in the new system's reliability.

4. **Integration with External Systems:** While the system is designed with an open API architecture, integrating with external third-party systems, such as Bank APIs or Donor Portals, may present technical challenges. These external systems may have strict security protocols or incompatible data formats that could require the development of custom middleware during the deployment phase.
5. **Technical Maintenance and Sustainability:** The sophisticated nature of the platform, particularly the Java/Spring Boot backend and AI microservices, requires a certain level of technical expertise to maintain. RPA may need to invest in upskilling its internal IT support or retaining a maintenance contract to ensure the server infrastructure, security protocols, and database optimizations remain effective after the project handover.

CONCLUSION

The **NGO Fund Platform** represents a transformative initiative for the Rwanda Paediatric Association (RPA), addressing critical financial compliance challenges while positioning the organization for sustainable growth in the increasingly competitive landscape of international aid. By integrating a centralized web-based command center with AI-driven predictive analytics and automated accountability workflows, this project directly targets the core institutional needs of enhancing funding transparency while optimizing internal reporting processes. The system's focus on digitizing manual ledgers, providing real-time budget insights, and creating a seamless "Donor-to-Beneficiary" audit trail will fundamentally reshape how RPA conducts its operations, elevating financial integrity while reducing administrative friction points.

Our implementation follows a structured **18-week Agile deployment plan** designed for rapid time-to-value. By prioritizing critical modules like **Donor Management** and **Fund Collection** in parallel development tracks, we ensure incremental delivery and early risk mitigation. This strategy allows RPA to realize immediate benefits while the system evolves, acknowledging the complexities of multi-donor compliance through a carefully sequenced rollout.

Upon successful implementation, the NGO Fund Platform will deliver measurable improvements across multiple operational dimensions, with projected enhancements including a **70% improvement in reporting accuracy**, a **40% reduction in audit preparation time**, and significant optimization of restricted fund utilization. Beyond these quantifiable metrics, the system will enable RPA to develop deeper strategic insights, create more resilient relationships with international partners, and make more informed decisions based on comprehensive financial data.

While acknowledging implementation challenges such as user adoption curves and initial data digitization efforts, the transformative potential of the NGO Fund Platform far outweighs these temporary limitations. The system establishes a technological foundation that not only addresses immediate operational needs but positions RPA as a model of non-profit excellence, directly contributing to **Rwanda's Vision 2050** goals for professionalizing the civil society sector and fostering accountability.

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APPENDICES

Case Study Organization Approval Letter



Office: 5 KK 361 st, Niboye, KICUKIRO
pediatric.rwanda@gmail.com
Phone number: 0789007495

Ref: 445/RPA/2025
Date: November 18th, 2025

To: Irakoze Tresor,

Adventist University of Central Africa (AUCA)

Ref: Approval for NGO fund platform project implementation

Dear Irakoze Tresor,

We acknowledge receipt of your letter dated September 06th, 2025, requesting permission to conduct research and implement an NGO Fund Platform system at the Rwanda Paediatric Association as part of your academic dissertation project.

After thorough consideration of your proposal, we are delighted to grant you approval to proceed with your project within our organization. Your proposed NGO Fund Platform demonstrates significant potential to improve our funding management processes and operational efficiency.

Approval Granted For:

- Collection of necessary data related to NGO funding operations
- Interviews with designated staff members
- Development and implementation of the NGO Fund Platform system
- Testing and validation of the system within our environment

We look forward to the successful completion of your project and believe it will bring mutual benefits to both your academic goals and our organizational development.

We extend our best wishes for the success of your project and your future professional career.

Sincerely,

Dr Jean Petit HABONIMANA
RPA – Executive Secretary

Current System Analysis

PART I

AS-IS PROCESS MODEL

Introduction

Introduction Non-Governmental Organizations (NGOs) in Rwanda, such as the Rwanda Paediatric Association (RPA), play a vital role in complementing government efforts to improve public health outcomes. Historically, these organizations have been supported by a combination of broad institutional grants and smaller, specific project-based funding. In the current operational landscape, the majority of financial compliance mechanisms are managed manually by finance officers using paper-based internal controls and disconnected Excel ledgers. While this traditional model supported the sector's initial phases of capacity building, Rwanda's rapid health sector transformation evidenced by the increasing complexity of multi-donor partnerships (USAID, UNICEF, WHO) has outpaced the system's ability to adapt. The increasing volume of restricted grants now demands a level of financial precision, traceability, and real-time reporting that analog systems can no longer sustain.

The current operational models in the national NGO funding landscape include **Unrestricted Funding Models** (general institutional support), **Fee-for-Service Models**, and **Restricted Grant-Based Models**. These models vary in scope and control: unrestricted funding allows for flexible allocation across administrative needs, whereas restricted grant models mandate that every franc be tied to a specific budget line item (e.g., "Malaria Workshop Logistics"). This study focuses specifically on the **Restricted Grant-Based Model**, which involves coordinating funds from multiple international donors, each with unique compliance rules. This model is the most significant for this study as it represents the greatest growth potential for Rwandan civil society but is currently the most impacted by the lack of digital infrastructure, serving as the primary source of revenue yet facing the highest audit risks.

The problem with this business model lies in its deep operational inefficiencies and inability to scale effectively. For example, RPA currently relies on manual spreadsheets and isolated physical documentation to manage its multifaceted grant portfolio, leading to significant information silos. Finance officers often record expenditure data on physical vouchers, which must be manually approved and then transcribed into Excel days later, introducing a high risk of

human error and data latency. As a result, the model suffers from unpredictable "burn rates," reporting delays, and weak internal control enforcement, as the Executive Director lacks real-time visibility into the organization's financial health. Furthermore, compliance with stringent international standards like **USAID Standard Provisions** is compromised when audit trails rely on fragmented paper receipts rather than digital logs. These challenges directly contradict Rwanda's **Vision 2050** goals of professionalizing the service sector, severely limiting the organization's ability to compete for larger direct funding opportunities.

In conclusion, while the manual grant management model has established RPA as a respected professional body, it is no longer viable in a rapidly evolving donor commitment landscape that demands 100% transparency and accountability. The absence of real-time budget data, automated compliance checks, and transparent audit trails hampers both operational sustainability and donor trust. To address this, there is an urgent need to design and implement a modern, web-based management platform, the **NGO Fund Platform**, that can improve transparency, optimize financial workflows, and align the civil society value chain with national digital transformation goals.

Modeling Of The Current System

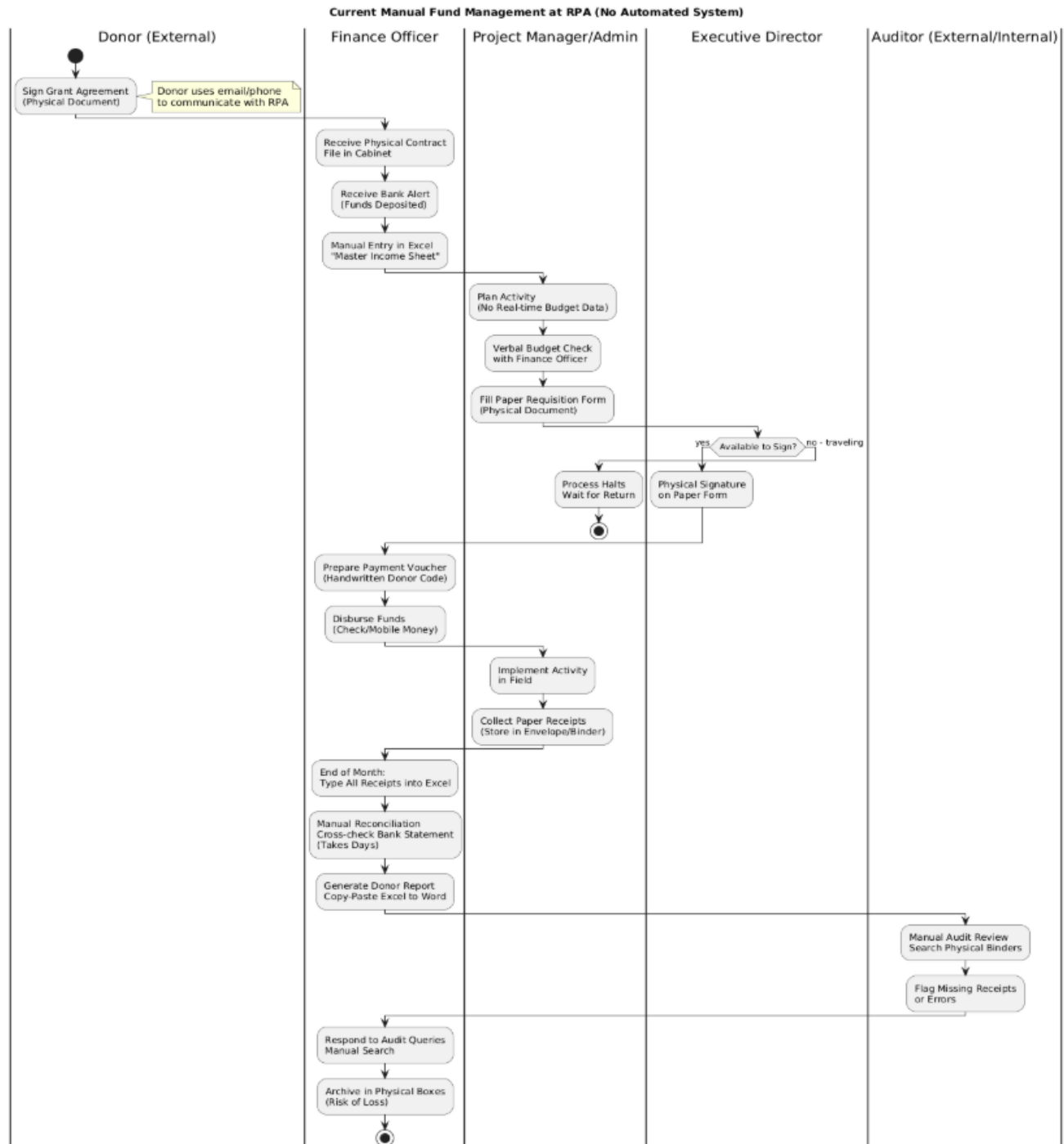


Figure 3: Current System

The following is a step-by-step overview of how the current manual fund management operations operate at RPA.

- **Step 1: Grant Agreement Signing:** The process begins when a donor (e.g., USAID, UNICEF) signs a grant agreement with RPA. This contract, detailing the budget and restricted line items, is stored physically in a filing cabinet, often disconnected from the finance team's daily view.
- **Step 2: Funds Receipt & Banking:** Funds are wired to the RPA bank account. The Finance Officer receives a bank alert and manually records the entry in a "Master Application Income" Excel sheet. This is the only record of "Cash In."
- **Step 3: Activity Planning & Budget Check:** A Project Manager plans an activity (e.g., a malaria workshop). Without a real-time system, they must verbally confirm available funds with the Finance Officer or rely on their own "shadow budget" spreadsheet, leading to frequent overspending risks.
- **Step 4: Requisition & Approval:** The manager fills out a physical "Requisition Form" and walks it to the Executive Director for a signature. If the Director is traveling, the process halts, delaying critical health interventions.
- **Step 5: Disbursement & Payment:** Once signed, the form returns to Finance. A check or mobile money transfer is prepared. The specific donor restriction code is often just a handwritten note on the payment voucher.
- **Step 6: Field Implementation & Receipt Collection:** Staff implement the activity and collect paper receipts for every expense (transport, meals). These small paper slips are kept in an envelope or binder until the end of the month.
- **Step 7: Manual Ledger Entry:** At the end of the month, the Finance Officer manually types every receipt detail into the Excel General Ledger. This duplicate data entry is a primary source of transcription errors and mismatched donor codes.
- **Step 8: Monthly Reconciliation:** The Finance Officer attempts to reconcile the manual Excel ledger with the bank statement. This process takes days of cross-referencing to find missing receipts or calculation errors.
- **Step 9: Reporting & Audit Preparation:** When a donor report is due, the Finance Officer must filter the Excel sheet for that specific donor's transactions, re-format the

data, and compile it into a Word document. This manual compilation is slow and prone to compliance flags.

- **Step 10: Feedback & Archiving:** After submission, donors may query specific expenses. Tracing a specific receipt back to a grant line item requires a laborious manual search through physical binders. Once resolved, records are archived in physical boxes, vulnerable to loss or damage.

PART II

PROBLEMS WITH THE CURRENT SYSTEM

Performance

Throughput

- The Finance team cannot handle multiple complex grants simultaneously. As the number of transactions increases, the backlog of unrecorded receipts grows.

Response Time

- Generating a financial status report for a specific project can take up to a week, delaying critical decision-making.

Information

- **Input:** High rate of human error during data entry (e.g., typing 10,000 instead of 100,000, or assigning the wrong donor code).
- **Output:** Reports are often static and outdated by the time they reach management. There is no real-time dashboard to see "Funds Remaining."
- **Data Integrity:** Excel sheets are often shared via USB or email, leading to version control conflicts (e.g., "Budget_Final_v3.xlsx").

Economics

- **Cost:** The hidden cost of staff time spent on manual data entry and error correction is significant.
- **Funding Risk:** The inability to produce timely, accurate audit trails poses a severe risk of having funding suspended by strict donors like USAID.

Control

- **Security:** Excel files are rarely password-protected effectively. Sensitive financial data is accessible to anyone who can access the finance computer.
- **Compliance:** There is no automated system to prevent spending "Restricted Funds" on unapproved categories. It relies entirely on the Finance Officer's memory of the grant rules.

Efficiency

- **Redundancy:** The same data is often written on a paper voucher, typed into an Excel log, and then typed again into a donor report.
- **Dependency:** The entire system relies on the knowledge of one or two staff members. If the Finance Officer leaves, the institutional memory of "which receipt belongs to which grant" is lost.

Service

- **Project Delays:** Program staff often waits days for fund disbursement approval because the physical signatory is out of the office.
- **Donor Satisfaction:** Delays in submitting financial reports can damage the relationship with key partners.